

An analysis of the interrelationship between maternal age, body mass index and racial origin in the development of gestational diabetes mellitus

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Objective To examine the individual association between advancing maternal age, body mass index (BMI) and racial origin with the development of gestational diabetes mellitus (GDM) and the interaction between these factors.

Design Retrospective study.

Setting Fifteen maternity units in northwest London between 1988 and 2000.

Population The study included 1688 women who developed GDM and 172 632 who did not. All women were nulliparous. BMI was calculated at first antenatal visit and maternal age and racial origin (White European, Black African, Black Caribbean or South Asian) were self-reported.

Methods Binary logistic regression analysis.

Main outcome measures Development of GDM within each racial group.

Results There was a strong positive association between advancing maternal age and increasing BMI, individually, and the development of GDM ($P < 0.01$ for both). Compared with White Europeans aged 20–24 years, the odds ratios for GDM development were significantly higher in women older than 30 years if they were White Europeans ($P < 0.001$), older than 25 years if they were Black Africans ($P < 0.001$) and older than 20 years if they were South Asians ($P < 0.001$). The odds ratios for GDM development were significantly higher in Black Africans and South Asians ($P < 0.001$ for both) irrespective of BMI, compared with White Europeans with normal BMI.

Conclusion Maternal age and BMI interact with racial group in relation to the prevalence of GDM. Both factors are important in the development of GDM, particularly so in Black African and South Asian women.

Keywords Body mass index, gestational diabetes mellitus, interrelationship, maternal age, maternal racial origin.

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Introduction

Gestational diabetes mellitus (GDM) complicates 3–5% of pregnancies and there is now good evidence that treatment for GDM reduces the frequency of adverse perinatal outcomes defined as death, bone fracture and nerve palsy.^{1,2} To allow effective treatment, early identification of GDM is essential. Currently in the UK, the National Institute for Health and Clinical Excellence (NICE)³ recommends a diagnostic test for GDM in women with traditional risk factors, in particular, body mass index (BMI) $>30 \text{ kg/m}^2$, previous history of GDM or delivery of a macrosomic baby, family

history of diabetes and racial origin with a high prevalence of diabetes. Although increasing maternal age is known to be a significant risk factor affecting the likelihood of GDM,^{4,5} it has not been included as one of the screening criteria because the secular increase in maternal age over recent years would have resulted in offering a diagnostic test to a high proportion of the pregnant population.⁶ Despite the wealth of evidence about the importance of the risk factors described above for the development of GDM, in particular older maternal age, high BMI and racial group, there is little published information about how these risk factors interact to influence an individual's risk of GDM.

The aim of the current study was to examine the individual association between maternal age, BMI and racial origin with the development of GDM, and their interaction.

Methods

Data were collected prospectively on 585 291 pregnancies in women attending for antenatal care and delivery at 15 maternity units in northwest London from 1988 to 2000, using the St Mary's Maternity Information System (SMMIS). The North West Thames Region had a population of 3.5 million, a wide variety of racial groups and included suburban, rural and metropolitan areas. Participating hospitals used SMMIS to collect data on every pregnancy cared for at their unit.

Data collection on 263 variables began at the first antenatal visit (before 16 weeks of gestation) and continued to 28 days postpartum and included demographic information, medical history, and information about the pregnancy, labour, delivery and outcome. The key features of SMMIS are that data were entered by trained clerks or midwives with on-line validation and prompting and standardised clinical definitions. The data have been extensively validated⁷ and the analysis of subsets has been reported in many papers to date.^{8,9}

At the first antenatal visit, BMI was calculated as weight (kg) (self-reported or measured by weighing if the woman did not know her weight), divided by height (m) squared. Racial groups were also classified by self-reporting. Initially the categories available were White European, Black African, Black Caribbean, South Asian, Oriental, Mediterranean and 'other racial group'. As the decade advanced, more categories were available, but they have subsequently been mapped to the same seven main groups to facilitate analysis. We chose to focus on White European, Black African, Black Caribbean and South Asian racial groups because these comprise the majority of the subjects in the data set. Information regarding diabetes in pregnancy was reported by the midwives at delivery and classified as no diabetes in pregnancy, pre-existing diabetes and diabetes that was diagnosed for the first time during the index pregnancy (GDM). At delivery, mean birthweight percentile calculation was based on percentiles derived from the Aberdeen survey, which take into consideration the gestational age at delivery and neonatal gender.¹⁰

Because we were interested specifically in GDM, we have excluded women with pre-existing diabetes ($n = 498$) from further analysis. We also excluded subjects with missing information regarding diabetes status during pregnancy ($n = 1348$), maternal age at booking ($n = 3758$), height ($n = 86\,058$), weight ($n = 20\,970$), or racial group ($n = 14\,378$) as well as those women giving birth before

24 weeks of gestation ($n = 899$) or with babies weighing <500 g ($n = 186$). We only studied first pregnancies to exclude any possible bias arising from including the same woman more than once and to eliminate any effect of parity. This resulted in one-third of the SMMIS database being appropriate to study.

Statistical analysis

Maternal characteristics and pregnancy outcomes in the two groups studied (non-GDM and GDM) were compared using the Student's *t* test and chi-square test for numerical and categorical data, respectively. Maternal age was divided into the following groups: below 20, 20–24, 25–29, 30–34, 35–39 and from 40 years of age. Maternal BMI was also divided according to the WHO international classification of BMI as follows: <18.5 (underweight), 18.50–24.99 (normal weight), 25.00–29.99 (overweight) and more or equal to 30.00 (obese).¹¹ Prevalence of GDM was calculated for each maternal age and BMI group.

Univariate analysis was initially used to examine the influence of maternal age and BMI on GDM development in each racial group. Binary logistical regression was used to develop statistical models for the development of GDM, including the risk variables of maternal age, BMI and race as well as the interactions between these variables. Models were constructed to examine the magnitude and significance of predictor variables on the prevalence of GDM. The results are presented as odds ratios (OR), 95% CI and *P*-values. White Europeans were used as the reference group for the analysis. All women were screened for GDM. However, as there was no single common screening method for GDM in the various maternity units, the effect of the unit on the reported prevalence of GDM was assessed. Preliminary analysis found no significant effect on the findings (data not shown but available on request) and so ORs have not been adjusted for unit. The statistical software package SPSS 17.0 (SPSS Inc., Chicago, IL, USA) was used for data analyses.

Figures were produced using binary logistic regression with GDM as the dependent variable, including the independent variables of maternal age and BMI and stratifying by racial group. Variables were tested using the Hosmer–Lemeshow test and then the values were transformed to find the best overall fit. The prevalence of GDM increased more rapidly with age than would be accounted for by a linear relationship and age squared produced the best overall fit.

Results

The study included 172 632 women who did not develop GDM (non-GDM group) and 1688 women who did (GDM group). A comparison between women who were included

in the study and those with missing information was also performed as part of the preliminary analysis. The prevalence of GDM in the missing population was not statistically different from that in the study population (1.3% versus 1%; $P = 0.446$). We therefore excluded women with missing information from further analysis because the selected study population adequately represented the population within the database.

The maternal characteristics and pregnancy outcomes of the groups studied are shown in Table 1. Compared with the non-diabetic group, women who developed GDM were more likely to be older, heavier, of non-white racial origin and to deliver large-for-gestational-age neonates. The rates of underweight, normal weight, overweight and obese women for each age group within the different racial groups are given in Table S1.

The prevalence of GDM in the different maternal age and BMI groups of our population is given in Table 2. There was a strong positive association between GDM development and advancing maternal age ($P < 0.001$) but this association varied markedly by racial group (Table 2, Figure 1). Using White European women age 20–24 years as a reference group, White European women older than 30 years had significantly higher ORs for GDM development. The ORs for GDM development were also significantly higher in the other racial groups but at a younger maternal age (older than 25 years if they were Black Africans or Black Caribbeans and older than 20 years if they were South Asians) (Table 3). White European women

under the age of 20 were the only racial group to have significantly lower ORs for developing GDM in comparison to the reference group. Similarly, there was a strong positive association between the prevalence of GDM and increasing maternal BMI in all the racial groups examined ($P < 0.001$) (Table 2, Figure 2). Using White European women with a normal BMI as the reference group, the ORs for GDM development were significantly higher in the overweight and obese White European and Black Caribbean groups and significantly higher in all BMI categories of Black African and South Asian women (Table 3).

The regression lines of the associations between maternal age squared, maternal BMI and the prevalence of GDM are shown in Figures 1 and 2. The regression lines were significantly different in all the non-White racial groups compared with White Europeans.

Interactions between variables

When the variable maternal age $\times$ BMI was included in the model for development of GDM, the interaction was not significant for any racial group ($P = 0.974$) (i.e. the influence of BMI does not vary with age). When the variable age $\times$ racial origin was included in the model, the interaction was significant for Black Africans ($P = 0.007$) and South Asians ($P = 0.005$) when compared with White Europeans. When the variable BMI $\times$ racial origin was included in the model, the interaction was significant only for the South Asians compared with White Europeans

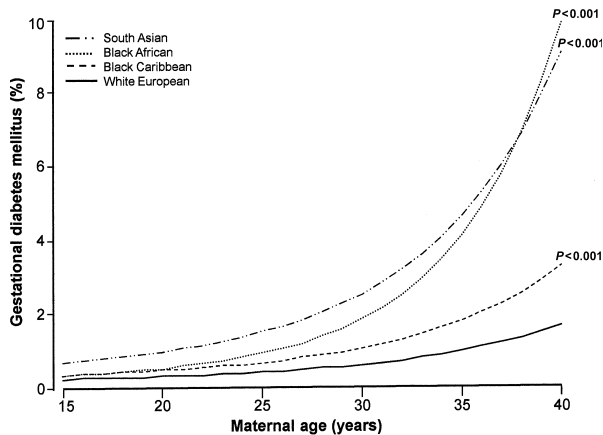
Table 1. Maternal demographic, clinical characteristics and pregnancy outcomes of the non-diabetic and gestational diabetic groups

Characteristic	Non-GDM (<i>n</i> = 172 632)	GDM (<i>n</i> = 1688)	<i>P</i> -value
Age (years)	26.8 $\pm$ 5.1	29.6 $\pm$ 5.2	<0.001
Gestational age at first antenatal visit (weeks)	13.0 (12.0–16.0)	13.0 (12.0–16.0)	0.567
Body mass index (kg/m²)	23.7 $\pm$ 4.0	26.6 $\pm$ 5.5	<0.001
Racial group, <i>n</i> (%)			
White European	130 325 (75.5)	876 (51.9)	<0.001
Black African	4840 (2.8)	87 (5.2)	
Black Caribbean	4648 (2.7)	50 (3.0)	
South Asian	19 685 (11.4)	401 (23.8)	
Oriental	2874 (1.7)	75 (4.4)	
Mediterranean	4390 (2.5)	79 (4.7)	
Other	5870 (3.4)	120 (7.1)	
Gestational age at delivery (weeks)	40.0 (39.0–41.0)	39.0 (38–40)	<0.001
Birthweight (g)	3523.8 $\pm$ 566.9	3305.9 $\pm$ 639.3	<0.001
Mean birthweight percentile	57.9 $\pm$ 28.2	63.9 $\pm$ 29.6	<0.001
Birthweight >90th percentile, <i>n</i> (%)	16 598 (9.6)	248 (14.6)	<0.001
Mode of delivery, <i>n</i> (%)			
Vaginal delivery	141 764 (82.1)	1101 (65.3)	<0.001
Caesarean section	30 868 (17.9)	587 (34.7)	

Values are expressed as mean $\pm$ standard deviation or as median (interquartile range).

Table 2. The rates of GDM and non-GDM in the study population according to the maternal age and body mass index groups within each different racial group

Variable	White European		Black African		Black Caribbean		South Asian	
	GDM n (%)	Non-GDM n (%)	GDM n (%)	Non-GDM n (%)	GDM n (%)	Non-GDM n (%)	GDM n (%)	Non-GDM n (%)
Age groups (years)								
<20	26 (0.2)	11 478 (99.8)	1 (0.3)	351 (99.7)	1 (0.1)	825 (99.9)	7 (0.5)	1541 (99.5)
20–24	134 (0.5)	28 559 (99.5)	10 (0.7)	1335 (99.3)	4 (0.3)	1320 (99.7)	77 (1.1)	7115 (98.9)
25–29	263 (0.5)	48 194 (99.5)	33 (1.6)	2064 (98.4)	22 (1.5)	1441 (98.5)	129 (1.8)	7133 (98.2)
30–34	308 (0.9)	32 167 (99.1)	26 (2.9)	882 (97.1)	20 (2.4)	814 (97.6)	125 (4.0)	3037 (96.0)
35–39	122 (1.4)	8128 (98.6)	12 (6.0)	187 (94.0)	2 (0.9)	217 (99.1)	50 (6.2)	750 (93.8)
≥40	23 (1.9)	1199 (98.1)	5 (21.7)	18 (78.3)	1 (3.1)	31 (96.9)	13 (11.4)	101 (88.6)
Body mass index groups								
Normal (18.5–24.9)	396 (0.5)	85 823 (99.5)	33 (1.2)	2729 (98.8)	15 (0.6)	2687 (99.4)	184 (1.4)	13 283 (98.6)
Overweight (25.0–29.9)	217 (0.8)	26 556 (99.2)	20 (1.6)	1234 (98.4)	17 (1.5)	1097 (98.5)	111 (3.4)	3122 (96.6)
Obese (≥30)	221 (2.1)	10 175 (97.9)	27 (5.6)	456 (94.4)	15 (2.6)	555 (97.4)	68 (7.4)	847 (92.6)

**Figure 1.** Frequency of gestational diabetes mellitus with increasing maternal age within each racial group. The maternal age is given as continuous variable. Regression lines for non-white racial groups are compared with White Europeans.

($P = 0.006$). These findings imply that maternal age is more important in the development of GDM in Black Africans and South Asians than in White Europeans and that BMI is more important in the South Asians compared with White Europeans.

Discussion

This retrospective study of prospectively collected data has demonstrated strong interactions between traditional risk factors such as advancing maternal age, increasing BMI and racial group on the development of GDM. Black African, Black Caribbean and South Asian pregnant women older than 25 years of age and even of normal BMI had a signifi-

cantly higher incidence of GDM compared with young White Europeans of normal BMI.

There is a wealth of literature confirming a strong positive correlation between racial origin, especially South Asians,^{12,13} obesity^{14–17} maternal age^{14,18} and the development of GDM, even after adjustment for potential confounders.¹⁸ However, most studies assessed the interaction between these risk factors as part of a mathematical model that was created to predict the development of GDM. Our study explicitly investigated the interrelationships of the aforementioned risk factors. We are not aware of any previous studies that have assessed the interaction between maternal racial origin and age on the development of GDM. Two previous studies have investigated the interrelationship between maternal racial origin and BMI. Ramos and Caughey¹⁹ reported that the incidence of GDM increased with increasing BMI among all racial groups. The incidence of GDM was highest in South Asian women in comparison to Latina, African American and White racial groups, suggesting that the influence of maternal weight on GDM development varies according to racial origin. Rodrigues et al.²⁰ found that, compared with non-native Canadians (mostly White Europeans), being a Cree North American was not a significant predictor of GDM in women of normal weight but the combination of both obesity and Cree racial origin significantly increased the risk of GDM, again suggesting an interaction between maternal race and BMI.

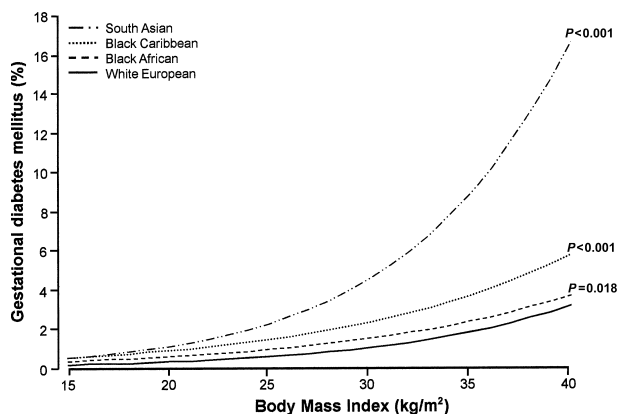
We have shown that, as far as the development of GDM is concerned, advancing maternal age is much more important in women of South Asian and Black African origin than in White Europeans. A non-white pregnant woman aged 25–29 years has a similar risk of developing GDM as

Table 3. Odds ratios (95% CI) for development of gestational diabetes mellitus according to advancing maternal age and increasing body mass index within each racial group

Variable	White European	Black African	Black Caribbean	South Asian
Age groups (years)				
<20	0.48 (0.31–0.73)*	0.60 (0.08–4.35)	0.25 (0.03–1.85)	0.96 (0.45–2.07)
20–24	1.00	1.59 (0.33–3.03)	0.64 (0.23–1.74)	2.30 (1.74–3.05)*
25–29	1.16 (0.94–1.43)	3.40 (2.32–5.00)*	3.25 (2.06–5.12)*	3.85 (3.02–4.91)*
30–34	2.04 (1.66–2.50)*	6.28 (4.10–9.61)*	5.23 (3.25–8.42)*	8.77 (6.85–11.22)*
35–39	2.97 (2.32–3.81)*	13.67 (7.44–25.11)*	1.96 (0.48–7.98)	14.05 (10.08–19.60)*
≥40	4.08 (2.61–6.38)*	59.20 (21.66–161.37)*	6.87 (0.93–50.72)	27.43 (15.02–50.08)*
Body mass index groups				
Normal (18.5–24.9)	1.00	2.62 (1.83–3.74)*	1.21 (0.72–2.02)	3.00 (2.51–3.57)*
Overweight (25.0–29.9)	1.77 (1.50–2.09)*	3.48 (2.21–5.48)*	3.35 (2.05–5.47)*	7.70 (6.22–9.53)*
Obese (≥30)	4.70 (3.98–5.55)*	12.83 (8.59–19.15)*	5.85 (3.47–9.87)*	17.39 (13.33–22.70)*

The reference groups were White European women of 20–24 years old and those with a normal body mass index (18.5–24.9).

* $P < 0.001$.

**Figure 2.** Frequency of gestational diabetes mellitus with increasing maternal body mass index within each racial group. The maternal body mass index is given as a continuous variable. Regression lines for non-white racial groups are compared with White Europeans.

a White European woman of 40 years; at least three to four times higher than a young (20–24 years old) White European. Advancing maternal age is also important in Black Caribbeans but we can only be confident about our results between 24 and 34 years old, because the number of such mothers in the older age groups was small. In our study we separated the Black racial group into Black Africans and Black Caribbeans because in the latter group the association of maternal age and BMI with development of GDM was less pronounced. A previous study has suggested that the incidence of GDM varies for subgroups within the Asian group.²¹ This could also be the case for Black populations and our study highlights the potential inaccuracies of grouping Black populations as a whole. The simi-

larities between White Europeans and Black Caribbeans may suggest a substantial white genetic admixture within this group, which perhaps influences future disease risk. However, this similarity may be a consequence of the relatively small number of cases in this racial group. Further research will be needed to clarify this issue.

It is not surprising that for the development of GDM, BMI appears to be of particular importance in South Asians and Black Africans. Compared with White Europeans, South Asians demonstrate increased propensity to insulin resistance, type 2 diabetes and cardiovascular disease, which has been attributed to the 'thrifty gene hypothesis' together with a clustering of metabolic abnormalities.^{22–25} South Asians have a greater predisposition to deposit intra-abdominal fat which is metabolically active and is strongly related to increased insulin resistance.^{26,27} Therefore, although the average BMI of a South Asian is lower than that of a White European, the central adiposity and its associated co-morbidity may be higher and likely to be underestimated. As regards Black African populations, studies outside the setting of pregnancy have clearly documented the higher prevalence of obesity, which is thought to be due, at least partly, to urbanisation and dramatic lifestyle changes, insulin resistance and type 2 diabetes.^{28,29}

The limitations of the study are that it was retrospective (although the data were collected prospectively) including 15 different maternity units in the North West Thames region and consequently there would have been some inconsistency in the screening methods and diagnosis of GDM. This may have led to either an under-reporting or over-reporting of GDM cases in some units and consequently the inclusion of some women with occult diabetes in the non-GDM group or the inclusion of normoglycae-

mic women in the GDM group. However, such misclassification would be expected to reduce the differences between groups and so the associations reported in the current study are likely, if anything, to be an underestimate of the true differences.

In summary, this study has demonstrated that there is a significant interaction between maternal age, BMI, racial origin and development of GDM. The finding that older age and higher BMI interact with racial group in relation to the prevalence of GDM emphasises the important public health message of healthy eating and weight control, which is particularly crucial in women of South Asian and Black African racial origin.

Conclusion

Advancing maternal age and BMI are more important risk factors for GDM in South Asian and Black African women than in White European or Black Caribbean women.

Disclosure of interests

None.

Contribution to authorship

MM formulated the hypothesis, performed the data analysis and wrote the first draft of the paper. MDS and PJS helped to formulate the hypothesis and participated in the analysis of the data and interpretation of the results. All authors have been involved in editing the first and final drafts of the paper and approving its contents.

Details of ethics approval

This research using St Mary's Maternity Information System database has been ethically approved by the St Mary's Local Research Ethics Committee, London.

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Supporting information

The following supplementary materials are available for this article:

Table S1. Rates of underweight (BMI < 18.5), normal weight (BMI 18.5–24.9), overweight (BMI 25.0–29.9) and obese (BMI ≥ 30) women within each racial group for each age group, irrespective of glycaemic status during pregnancy.

Additional Supporting Information may be found in the online version of this article.

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